9.1 The Pythagorean Theorem

Name _____ Period ____ Date _____

Geometry 9.1 · Student sheet

I can…

- use the Pythagorean Theorem to find a missing side of a right triangle.
- tell whether three lengths form a Pythagorean triple.
- use the converse of the Pythagorean Theorem to decide whether a triangle is right, acute, or obtuse.

Vocabulary · fill in the missing word

Hypotenuse — the side _____ from the right angle; the longest side of a right triangle

Legs — the two sides that form the _____ angle

Pythagorean triple — three _____ integers a, b, and c with $c^2 = a^2 + b^2$

Converse of the Pythagorean Theorem — if $c^2 = a^2 + b^2$, then the _____ is a right triangle

Part 1 · Discovery: is it a right triangle?

Partner A draws each triangle (longest side as the base) and measures the **largest angle**. Partner B finds $a^2 + b^2$ and c^2, where c is the longest side.

Sides	$a^2 + b^2$	c^2	Largest angle	Right, acute, or obtuse?
3, 4, 5				
5, 12, 13				
6, 8, 10				
4, 5, 6				
3, 4, 6				

When $a^2 + b^2 = c^2$, the largest angle is _____.

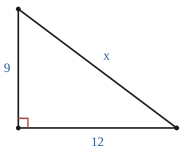
Part 2 · Notes and examples (fill in each note, then work the example)

Pythagorean Theorem: in a right triangle, the square of the length of the _____ equals the sum of the squares of the lengths of the _____: $c^2 = a^2 + b^2$.

Pythagorean triple: three positive _____ a, b, c with $c^2 = a^2 + b^2$. Common ones: 3, 4, 5; 5, 12, 13; 8, 15, 17; 7, 24, 25.

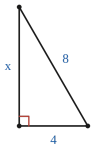
Example 1

Find the value of x. Then tell whether the side lengths form a Pythagorean triple.



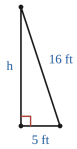
Example 2

Find the value of x. Then tell whether the side lengths form a Pythagorean triple.



Example 3

A 16-foot ladder leans against a wall. The bottom of the ladder is 5 feet from the wall. How high up the wall does the ladder reach? Round to the nearest tenth.



Converse of the Pythagorean Theorem: if $c^2 = a^2 + b^2$ (c the longest side), then the triangle is a _____ triangle.

Example 4

Tell whether each triangle is a right triangle.

a. side lengths 10, 24, 26

b. side lengths 8, 10, 13

Acute or obtuse: with c the longest side, if $c^2 < a^2 + b^2$ the triangle is _____; if $c^2 > a^2 + b^2$ it is _____.

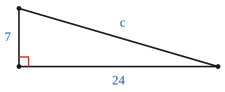
Example 5

Verify that segments with lengths 4.3, 5.2, and 6.1 form a triangle. Is the triangle acute, right, or obtuse?

Part 3 · Partner practice

Partner 1

The legs of a right triangle are 7 and 24. Find the hypotenuse. Do the side lengths form a Pythagorean triple?



Partner 2

A triangle has side lengths 6, 8, and 11. Is it acute, right, or obtuse?